INCH-POUND

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DETAIL SPECIFICATION

ELECTRODES AND RODS – WELDING, BARE, NICKEL, NICKEL ALLOY, COPPER-NICKEL ALLOY

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nickel, nickel alloy, and copper-nickel alloy bare welding electrodes and rods for use with the submerged arc welding (SAW) (when used with a neutral flux), gas metal-arc welding (GMAW), and gas tungsten-arc welding (GTAW) welding processes.

1.2 Classification. Electrodes and rods are furnished in the following classes, types, and sizes as specified (see 6.2).

1.2.1 Classes.

a. Class 1 – Wound bare electrode (wire form) (see table I for types and sizes).
   (1) 1a – 25-pound spool weight.
   (2) 1b – 2-pound spool weight (furnished in 0.035 inch only).
   (3) 1c – 100-pound drum weight.
   (4) 1d – 200-pound drum weight.
   (5) 1e – 300-pound drum weight.
   (6) 1f – 400-pound drum weight.
   (7) 1g – 500-pound drum weight.
   (8) 1h – 600-pound drum weight.
   (9) 1i – 700-pound drum weight.

b. Class 2 – Cut length rods (see table II for types and sizes).

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AMSC N/A FSC 3439

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TABLE I. Electrodes, class 1, types and sizes. [1/]

Types	Alloy	Diameter size (inch)
MIL-EN52i	Ni-Cr-Fe	0.035, 0.045, 0.062
MIL-EN60	Ni-Cu	
MIL-EN61	Ni	
MIL-EN62	Ni-Cr-Fe	
MIL-EN64	Ni-Cu-Al	
MIL-EN67	Cu-Ni (70-30)	
MIL-EN6A	Ni-Cr-Fe-Ti-Mn	
MIL-EN82	Ni-Cr-Mn-Nb	
MIL-EN82H	Ni-Cr-Mn-Nb	
MIL-EN6N	Ni-Cr (80-20)	
MIL-EN625	Ni-Cr-Mo-Nb	

NOTE:

[1/] Unless otherwise specified (see 6.2), sizes are as specified in table I.

TABLE II. Rods, class 2, types and sizes. [1/]

Types	Alloy	Diameter size (inch)
MIL-RN52i	Ni-Cr-Fe	0.035, 0.045, 1/16, 5/64, 3/32, 1/8, 5/32, 3/16, 1/4
MIL-RN60	Ni-Cu	
MIL-RN61	Ni	
MIL-RN62	Ni-Cr-Fe	
MIL-RN64	Ni-Cu-Al	
MIL-RN65	Ni-Fe-Cr-Mo-Cu	
MIL-RN67	Cu-Ni (70-30)	
MIL-RN69	Ni-Cr-Fe-Ti-Al-Nb	
MIL-RN6A	Ni-Cr-Fe-Ti-Mn	
MIL-RN82	Ni-Cr-Mn-Nb	
MIL-RN82H	Ni-Cr-Mn-Nb	
MIL-RN6N	Ni-Cr (80-20)	
MIL-RN625	Ni-Cr-Mo-Nb	

NOTE:

[1/] Unless otherwise specified (see 6.2), sizes are as specified in table II.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-10430 - Welding Rods and Electrodes; Packaging of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org/.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.01M/A5.01 - Welding and Brazing Consumables – Procurement of Filler Metals and Fluxes

AWS A5.02/A5.02M - Specification for Filler Metal Standard Sizes, Packaging, and Physical Attributes

AWS A5.32M/A5.32 - Welding Consumables – Gases and Gas Mixtures for Fusion Welding and Allied Processes

AWS B4.0 - Standard Methods for Mechanical Testing of Welds

(Copies of these documents are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM B167 - Standard Specification for Nickel-Chromium-Aluminum Alloys, Nickel-Chromium-Iron Alloys, Nickel-Chromium-Cobalt-Molybdenum Alloy, Nickel-Iron-Chromium-Tungsten Alloy, and Nickel-Chromium-Molybdenum-Copper Alloy Seamless Pipe and Tube

ASTM B168 - Standard Specification for Nickel-Chromium-Aluminum Alloys (UNS N06699), Nickel-Chromium-Iron Alloys (UNS N06600, N06601, N06603, N06690, N06693, N06025, N06045, and N06696), Nickel-Chromium-Cobalt-Molybdenum Alloy (UNS N06617), Nickel-Iron-Chromium-Tungsten Alloy (UNS N06674), and Nickel-Chromium-Molybdenum-Copper Alloy (UNS N06235) Plate, Sheet, and Strip

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ASTM D635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

ASTM E354 - Standard Methods for Chemical Analysis of High-Temperature, Electrical, Magnetic, and Other Similar Iron, Nickel, and Cobalt Alloys

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60974 - Arc Welding Equipment

(Copies of this document are available online at www.iec.ch.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org/home.html).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA EW 1 - Electric Arc-Welding Power Sources

(Copies of this document are available online at www.nema.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The electrodes and rods furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials. Materials shall be as specified in 3.2.1 and 3.2.2.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Material from previous production runs. Materials from previous production runs shall be excluded from any new production run to be offered for acceptance, unless the materials have been determined to be acceptable for meeting all requirements of this specification (see 4.5.2.1).

3.3 General requirements. General requirements shall be as specified in 3.3.1 through 3.3.5.2.

3.3.1 Finish. Electrodes and rods shall have a smooth finish that is free from slivers, depressions, scratches, seams, laps, scale, and foreign matter that would adversely affect the welding characteristics, the operation of the equipment, or the properties of the deposited metal (see 4.5.2.2.1).

3.3.2 Chemical composition. The chemical composition of electrodes (wire form) and rods shall be as specified in table III (see 4.5.2.3).

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TABLE III. Chemical composition percent of electrodes and rods. [1/, 2/]

Types MIL-	C [3/]	Mn	Fe	S [4/]	Si	Cu	Ni plus Co	Co	Al	Ti	Cr	Nb plus Ta	Ta	Mo	P	Pb	N	Mg	Other elements total [5/]
EN52i; RN52i	0.03-0.05	2.50-3.00	2.00-3.00	0.002	0.01-0.10	0.10	Balance	0.1	0.20	0.20-0.40	26.0-28.0	2.20-2.40	0.30	0.010	0.008	0.010	0.015 - 0.025	0.001 – 0.010	0.50 [6/]
EN60; RN60	0.15	4.00	2.50	0.015	1.25	Balance	62.0 - 69.0	--	1.25	1.5 - 3.0	--	--	--	--	0.020	0.010	--	--	0.50
EN61; RN61	0.15	1.00	1.00	0.015	0.75	0.25	93.0 min.	--	1.50	2.0 - 3.5	--	--	--	--	0.030	0.010	--	--	0.50
EN62; RN62	0.08	1.00	6.0 - 10.0	0.015	0.35	0.50	70.0 min.	0.1	--	--	14.0 - 17.0	1.5 - 3.0	0.30	--	0.030	0.010	--	--	1.0
EN64; RN64	0.25	1.50	2.00	0.015	1.00	Balance	63.0 - 70.0	--	2.0 - 4.0	0.25 - 1.00	--	--	--	--	0.030	0.010	--	--	0.50
RN65	0.05	1.00	22.0 min.	0.03	0.50	1.5 - 3.0	38.0 - 46.0	--	0.20	0.60 - 1.2	19.5 - 23.5	--	--	2.5 - 3.5	0.030	0.010	--	--	0.50
EN67; RN67	0.04	1.00	0.4 - 0.7	0.015	0.25	Balance	29.0 - 32.0	--	--	0.2 - 0.5	--	--	--	--	0.020	0.010	--	--	0.50
RN69	0.08	1.00	5.0 - 9.0	0.015	0.50	0.50	70.0 min.	--	0.4 - 1.0	2.00 - 2.75	14.0 - 17.0	0.70 - 1.20	0.30	--	0.030	0.010	--	--	0.50
EN6A; RN6A	0.08	2.00 - 2.75	8.00	0.015	0.35	0.50	67.0 min.	--	--	2.5 - 3.5	14.0 - 17.0	--	--	--	0.030	0.010	--	--	0.50
EN82; RN82	0.10	2.50 - 3.50	3.00	0.015	0.50	0.50	67.0 min.	0.1	--	0.75	18.0 - 22.0	2.0 - 3.0	0.30	--	0.030	0.010	--	--	0.50
EN82H; RN82H	0.030 - 0.10	2.50 - 3.50	3.00	0.015	0.50	0.50	67.0 min.	0.1	--	0.75	18.0 - 22.0	2.0 - 3.0	0.30	--	0.030	0.010	--	--	0.50
EN6N; RN6N	0.08 - 0.15	1.00	2.00	0.015	0.30	0.50	75.0 min.	--	0.40	0.15 - 0.50	19.0 - 21.0	--	--	--	0.030	0.010	--	--	0.50
EN625; RN625	0.10	0.50	5.00	0.015	0.50	1.00	58.0 min.	0.1	0.40	0.40	20.0 - 23.0	3.15 - 4.15	0.30	8.0 - 10.0	0.020	0.010	--	--	0.50

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TABLE III. Chemical composition percent of electrodes and rods – Continued.

NOTES:

1/ Analysis shall be made for the elements for which specific values are shown. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not present in excess of the limits specified for "other elements, total".

2/ Single values shown are maximum percentages, except where otherwise specified.

3/ When specified (see 6.2), a minimum carbon content shall be required.

4/ When specified (see 6.2), sulfur shall not exceed 0.007 percent.

5/ Includes lead, tin, and zinc.

6/ No intentional additions of unspecified elements.

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3.3.3 Mechanical properties. The deposited weld metal shall exhibit the mechanical properties as specified in table IV (see 4.5.2.4).

TABLE IV. Mechanical properties of welds.

Types MIL-	Tensile strength 1/, 5/	Elongation in 4X dia.	Ductility (full capacity of jig)	Fissures per bend specimen
	lb/in^2 minimum	Percent minimum		
EN52i; RN52i	80,000	25	Side bends (2)	4/
EN60; RN60	70,000	30	Side bends (2)	4/
EN61; RN61	60,000	20	--	
EN62; RN62	80,000	30	Side bends (2)	4/
EN64; RN64	100,000 2/	15	--	--
RN65	80,000	25	--	--
EN67; RN67	50,000	30	Side bends (2)	4/
RN69	125,000 3/	5	--	--
EN6A; RN6A	80,000	30	Side bends (2)	4/
EN82; RN82	80,000	30	Side bends (2)	4/
EN82H; RN82H	80,000	30	Side bends (2)	4/
EN6N; RN6N	80,000	20	--	--
EN625; RN625	110,000	30	Side bends (2)	4/

NOTES:

1/ As welded properties.

2/ Charge welded plate into a furnace which has been preheated to 1,450 degrees Fahrenheit (°F), bring to a steady state and hold at temperature for 30 minutes and air cool or water quench. Age harden at 1,100 ±25 °F for 16 hours and furnace at approximately 25 °F per hour down to 900 °F and air cool to room temperature.

3/ Stress relief at 1,800 ±25 °F for 30 minutes. Age harden at 1,300 °F for 20 hours and air cool to room temperature.

4/ The presence in either specimen of (a) a single tear, fissure, crack, or non-metallic having a dimension larger than 3/32 inch measured in any direction, (b) more than one fissure from 1/16 inch to 3/32 inch, inclusive, measured in any direction, (c) more than six fissures visible to the unaided eye 1/16 inch or less measured in any direction, or (d) more than six fissures total, all sizes, visible to the unaided eye shall be cause for rejection, subject to the retest requirements of 4.5.3.

5/ Two-tenths percent offset yield strength shall be reported for information.

3.3.4 Soundness. The groove welds shall meet the requirements in accordance with class I of MIL-STD-2035 (see 4.5.2.5).

3.3.5 Alloy identity. Alloy identity shall be as specified in 3.3.5.1 and 3.3.5.2.

3.3.5.1 All phases of production prior to final marking. The material shall be of the type specified when tested in accordance with 4.5.2.6 during all phases of production prior to final marking.

3.3.5.2 After final marking. When specified (see 6.2), each electrode and rod shall be of the material type specified when tested in accordance with 4.5.2.6 after final marking. Identification of each spool, coil, or container of cut length rods tested after final marking and meeting the acceptance criteria shall include the words “Alloy tested”.

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3.4 Class 1 – wound electrode. Class 1 wound electrodes shall be as specified in 3.4.1 through 3.4.4.2.

3.4.1 Tolerance (diameter size). The diameter of the electrode shall not vary from the nominal diameter by more than the limits identified in AWS A5.02/A5.02M (see 4.5.2.2.1).

3.4.2 Weight of wound electrode. The nominal net weight of the wound electrode shall be as follows:

a. Class la – 25 pounds ±20 percent. (Up to 20 percent of any lot of spools may contain 12 to 20 pounds net weight) (see 4.5.2.2.2).

b. Class lb – 2 pounds ±10 percent (see 4.5.2.2.2).

c. Class 1c through 1i – 100 to 700 pounds, tolerance shall be as specified (see 4.5.2.2.2 and 6.2).

3.4.3 Winding. Each spool shall contain an electrode of one continuous length made from a single heat of material. The electrode shall be closely wound in layers in such a manner to avoid producing kinks, waves, or sharp bends, and shall be free to unwind without restriction caused by overlapping or wedging. The outside end of the wound electrode shall be securely fastened at a readily located point. The starting end of the spooled and coiled electrode shall be firmly fastened so that it is readily visible to the welder or welding operator, and both ends of the electrode shall be accessible for inspection purposes (see 4.5.2.2.1).

3.4.3.1 Cast and helix.

3.4.3.1.1 Class 1a cast. The cast of the unrestrained circle formed using the procedure as specified in 4.5.2.7.1.1 shall be not less than 15 inches and not greater than 50 inches in diameter.

3.4.3.1.2 Class 1a helix. The electrode shall not exhibit a tendency toward spiraling that interferes with welding performance. The vertical separation between turns of the released specimen shall not exceed 1 inch when using the procedure as specified in 4.5.2.7.1.1.

3.4.3.1.3 Temper –class 1b. The electrode shall possess a temper that assures uniform feeding in welding equipment (see 4.5.2.7.2).

3.4.3.1.4 Classes 1c through 1i. The wire shall be suitable for uniform, uninterrupted feeding in automatic and semi-automatic welding equipment (see 4.5.2.7.3).

3.4.4 Electrode identification. Identification shall be as specified in 3.4.4.1 and 3.4.4.2.

3.4.4.1 Classes 1a and b. MIL-type, size, lot or control number and date of processing (cleaning and spooling) shall be shown on the spool flange exposed to view when loaded in the equipment (see 4.5.2.2.1).

3.4.4.2 Classes 1c through 1i. Tags made of material that will resist effacement or destruction and bearing the specification number, MIL-type, size, date of processing, and lot control number shall be firmly attached at the inner end of the coiled filler metal. Coils wound on supports shall be identified with the above information placed directly on the support member in such a manner as not to be readily removable. Filler metal wound in drums shall be identified with the above information placed directly on the drum in such a manner as not to be readily removable (see 4.5.2.2.1).

3.5 Class 2 – cut length rods. Class 2 cut length rods shall be as specified in 3.5.1 through 3.5.3.

3.5.1 Tolerance (diameter size). The diameter of the rods shall not vary from the nominal diameter by more than the limits in accordance with AWS A5.02/A5.02M (see 4.5.2.2.1).

3.5.2 Length. The length of the rods shall be 12, 18, or 36 inches, plus 0, minus ½ inch, as specified (see 6.2). In the case of 36-inch rods, up to 10 percent of the rods in any one container may be shorter than 36 inches, but not shorter than 24 inches (see 4.5.2.2.1).

3.5.3 Identification. Cut-length rods shall be identified by positive and legible methods such as imprinting or indenting the applicable type designation number (see table II) at one or more locations on the rod surface approximately 1 inch from rod end, or shall be identified by pressure-sensitive, plastic-coated tape imprinted with applicable type number at one or more locations and attached to the rod 1 inch from its end. Imprints on rods or on tape shall be with fade-proof ink and shall be resistant to oils, solvents, and all atmospheric conditions and to normal wear and tear encountered in shipping and handling. The MIL-type, size, lot or control number, and date of processing shall be marked on interior packages (see 4.5.2.2.1).

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3.6 Plastic spool burn characteristics. When specified (see 6.2), plastic spools shall not burn or shall stop burning in accordance with 4.5.2.8.

3.7 Container requirements. Container requirements shall be as specified in 3.7.1 through 3.7.3.3.

3.7.1 Class 3 spools, class 4 drums, and class 6 containers. The electrodes and rods shall be furnished on class 3 spools, class 4 drums, and class 6 containers, as applicable, and shall be preserved at level A, C, or commercial, packed at level A, B, C, or commercial in accordance with MIL-W-10430, or in accordance with AWS A5.02/A5.02M, as specified (see 6.2). Additional requirements shall be as specified (see 6.2 and 4.5.2.2.1).

3.7.2 Alternate class 6 container. In lieu of the class 6 container in accordance with MIL-W-10430, rods of like type and diameter may be packaged in 10-pound spiral-wound fiber tubes from 0.125- to 0.135-inch thick containing three layers of 0.007- to 0.008-inch thick waterproof duplex kraft wound with a ¼-inch overlap. End closures shall be snug fitting flexible polyethylene caps reinforced and sealed with waterproof tape (see 4.5.2.2.1).

3.7.3 Container marking requirements. Markings for electrodes and rods shall be in accordance with 3.7.3.1 through 3.7.3.3 and MIL-W-10430. Additional requirements shall be as specified (see 6.2 and 4.5.2.2.1).

3.7.3.1 Coils with liners, spools, drums with and without support. In addition to special marking in accordance with MIL-W-10430 on warning labels, each coil with a liner, spool, or drum shall have the MIL-W-10430 specified warning label, or equivalent, permanently affixed to the body of the coil, spool, or drum in a prominent position and in legible type. Coils without support shall have a tag bearing the MIL-W-10430 specified warning label, or equivalent, in legible type securely attached to both ends of the coil (see 4.5.2.2.1).

3.7.3.2 Asbestos-free. Each unit container shall be asbestos-free. Each unit container shall be marked “Asbestos-Free” (see 4.5.2.2.1).

3.7.3.3 Lot identification. Lot identification shall be accomplished by marking each unit and shipping container with a unique manufacturer’s control number that is traceable to the original manufacturer’s records. If these control numbers are encoded, the interpretive key shall be furnished to the consignee. The control number shall be in a position that can be easily located by the purchaser. This control number shall be in a position that permits ready location by the consignee (see 4.5.2.2.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (see tables V and VI):

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

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TABLE V. Summary of required examinations for qualification and conformance inspections.

Examination	Type MIL-	Class	Size	Sampling	Requirement	Verification	Qualification	Conformance
Material from previous production runs	All	All	All	4.5.2.1	3.2.22	4.5.2.1	--	X
Finish		See 4.2	See 4.2	4.3.1.3	3.3.1	4.5.2.2.1	X	--
Finish		All	All		3.3.1	4.5.2.2.1	--	X
Tolerance (diameter size)		Class 1	See 4.2		3.4.1	4.5.2.2.1	X	--
Tolerance (diameter size)		Class 1	All		3.4.1	4.5.2.2.1	--	X
Weight of wound electrode		Class 1	All		3.4.2	4.5.2.2.2	--	X
Winding		Class 1	All		3.4.3	4.5.2.2.1	--	X
Electrode Identification		Class 1	All		3.4.4	4.5.2.2.1	--	X
Tolerance (diameter size)		Class 2	See 4.2		3.5.1	4.5.2.2.1	X	--
Tolerance (diameter size)		Class 2	All		3.5.1	4.5.2.2.1	--	X
Cut length rod length		Class 2	See 4.2		3.5.2	4.5.2.2.1	X	--
Cut length rod length		Class 2	All		3.5.2	4.5.2.2.1	--	X
Cut length rod Identification		Class 2	All		3.5.3	4.5.2.2.1	--	X
Containers and marking.		Class 1 and 2, as applicable	All	4.3.1.8	3.7	4.5.2.2.1	--	X

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TABLE VI. Summary of required tests for qualification (Qual.) and conformance (Conf.) inspections. 1/, 2/, 3/

Type MIL-	Class	Size	Sampling	Req'ts	Verification	Cast	Helix	RT	Tensile	Side Bend	Chemical	Metallurgical	Alloy identity	Qual.	Conf.
All	1a and 1c thru 1i	See 4.2	4.3.1.4	3.4.3.1	4.5.2.7	X	X	--	--	--	--	--	--	X	--
All	1a and 1c thru 1i	See 4.2	4.3.1.7	3.3.4	4.5.2.5	--	--	X	--	--	--	--	--	X	--
All	1a and 1c thru 1i	See 4.2	4.3.1.7	table IV	4.5.2.4.1	--	--	--	X	--	--	--	--	X	--
EN52i; RN52i	1a and 1c thru 1i	See 4.2	4.3.1.7	table IV	4.5.2.4.2 table IV	--	--	--	--	X	--	Fissure examination	--	X	--
EN60; RN60															
EN62; RN62															
EN67; RN67															
EN6A; RN6A															
EN82; RN82															
EN82H; RN82H															
EN625; RN625															
All	1a and 1c thru 1i	See 4.2	4.3.1.6	table III	4.5.2.3	--	--	--	--	--	X	--	--	X	--
All	1a and 1c thru 1i	See 4.2	4.5.2.6	3.3.5	4.5.2.6	--	--	--	--	--	--	--	X	X	--

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TABLE VI. Summary of required tests for qualification (Qual.) and conformance (Conf.) inspections 1/, 2/, 3/ - Continued.

Type MIL-	Class	Size	Sampling	Req'ts	Verification	Cast	Helix	RT	Tensile	Side Bend	Chemical	Metallurgical	Alloy identity	Qual.	Conf.
All	All	All	4.3.1.4	3.4.3.1	4.5.2.7	X	X	--	--	--	--	--	--	--	X
All	All	All	4.3.1.7	3.3.4	4.5.2.5	--	--	X	--	--	--	--	--	--	X
All	All	All	4.3.1.7	table IV	4.5.2.4.1	--	--	--	X	--	--	--	--	--	X
EN52i; RN52i	All	All	4.3.1.7	table IV	4.5.2.4.2 table IV	--	--	--	--	X	--	Fissure examination	--	--	X
EN60; RN60															
EN62; RN62															
EN67; RN67															
EN6A; RN6A															
EN82; RN82															
EN82H; RN82H															
EN625; RN625															

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TABLE VI. Summary of required tests for qualification (Qual.) and conformance (Conf.) inspections [1/, 2/, 3/] - Continued.

Type MIL-	Class	Size	Sampling	Req'ts	Verification	Cast	Helix	RT	Tensile	Side Bend	Chemical	Metallurgical	Alloy identity	Qual.	Conf.
All	All	All	4.3.1.6	table III	4.5.2.3	--	--	--	--	--	X	--	--	--	X
All	All	All	4.5.2.6	3.3.5	4.5.2.6	--	--	--	--	--	--	--	X	--	X

NOTES:

1/ A certificate of conformance is required (see 4.3.2).

2/ Plastic spool burn characteristics, sampling, and verification requirements are specified in 3.6, 4.3.1.9, and 4.5.2.8, respectively. These requirements apply to all class 1 electrodes for conformance inspection.

3/ Where a heat of metal is processed into both class 1 electrodes and class 2 rods of the same size and sample selected from class 1 and passes mechanical and radiographic tests, it will not be necessary to perform these tests using class 2 rods of the same size. Where a heat of metal is processed to class 2 rods only, the required mechanical and radiographic tests shall be performed on welds made with GTAW.

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4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the Naval Sea Systems Command (NAVSEA). Qualification inspection shall include the examinations and tests specified in 4.4.

4.2.1 Class 1 and class 2. Where qualification is desired for the alloy types furnished in both class 1 and class 2 (see tables I and II), tests and examinations shall be conducted on class la, 1/16-inch diameter electrodes of the particular type alloy, and qualification will include both class 1 electrodes and class 2 rods (except as specified in 4.2.2) and all sizes for the particular alloy involved, if production cleaning methods are the same for all classes and sizes.

4.2.2 Class 2 rods only. Where qualification is desired for alloy types (see table II) furnished in class 2 rods only (RN65 and RN69), tests and examinations shall be conducted on 1/16-inch diameter rods and qualification will include all other sizes of the class and type alloy involved. Welds shall be made with GTAW.

4.2.3 Samples for qualification tests. One spool of class la, 1/16-inch diameter electrodes or a quantity of class 2 rods sufficient for tests, in accordance with 4.2 or 4.2.2, as applicable, shall be selected from production runs in the presence of the Government representative for transmittal to the testing laboratory.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests specified in 4.4.

4.3.1 Sampling for conformance inspection.

4.3.1.1 Lot. For the purposes of sampling and inspection, a lot of rod or electrode is defined as the quantity of one type and size produced from the same heat of metal. The lot size shall be expressed in pounds of rod or pounds of electrode.

4.3.1.2 Sampling for inspection of filled containers. Unit packages and shipping containers shall be sampled and inspected in accordance with MIL-W-10430.

4.3.1.3 Sampling for examination of rods and electrode. These sampling requirements apply to the examination requirements of 4.5.2.2. Sample rods and electrodes from each lot shall be selected either from the production line immediately prior to packaging or from filled unit packages. If selected from the production line, the total sample shall be in accordance with 4.3.1.5 and the electrodes or rods shall be selected throughout the run so that all parts of the run are represented. If selection is made after the packaging operation, the total sample shall be in accordance with 4.3.1.5, and approximately the same number of electrodes or rods shall be selected from each of the sample unit packages five cut lengths (rods) of each 5- or 10-pound package, 3 feet of each 2-pound spool, and an electrode length that will form one complete circle shall be taken from each spool 25 pounds (nominal) or larger.

4.3.1.3.1 Non-conforming material. Any sample coil, spool, or box of rods that contains any nonconforming material shall be counted as defective. If in any sample, the number of coils, spools, or boxes that do not conform to this specification exceeds the acceptance number determined in 4.3.1.5, this shall be cause for rejection of the lot, except as permitted in 4.3.1.4 (see 6.11). The retest requirements of 4.5.3 apply except where prohibited, as specified in 4.5.3.1.

4.3.1.4 Sampling for cast and helix tests. Sampling for cast and helix texts shall be in accordance with 4.3.1.4.1 and 4.3.1.4.2.

4.3.1.4.1 Class 1a. Specimens of class la electrode shall be cut from the spools selected in accordance with 4.3.1.5. (see 4.5.2.7). If any specimen fails either test, both tests shall be performed on every spool. The retest requirements of 4.5.3 apply for tests performed on any individual spool, except that the entire lot need not be rejected upon failing a retest. Only the spool shall be rejected.

4.3.1.4.2 Classes 1b through 1i. The number of spools and coils of classes 1b through 1i for determining conformance to 3.4.3.1.3 or 3.4.3.1.4, as applicable, shall be selected in accordance with 4.3.1.5. If any test fails to meet the requirements of 3.4.3.1.3 or 3.4.3.1.4 using the manufacturer's procedure (see 4.5.2.7.2 and 4.5.2.7.3), then the tests shall be performed on every spool or drum, as applicable. The retest requirements of 4.5.3 apply for tests performed on any individual spool or drum, except that the entire lot need not be rejected upon failing a retest. Only the spool or drum shall be rejected.

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4.3.1.5 Inspection levels. Sampling for examination of all classes of electrodes and cut rods shall be in accordance with the procedures of ANSI/ASQ Z1.4 The percent defective for determining the acceptance level using the procedures of ANSI/ASQ Z1.4 shall be as specified (see 6.2 and 6.7). The lot size shall be in terms of pounds whereas the sample size shall be in number of spools, packages, or drums. The inspection levels for each class shall be in accordance with table VII.

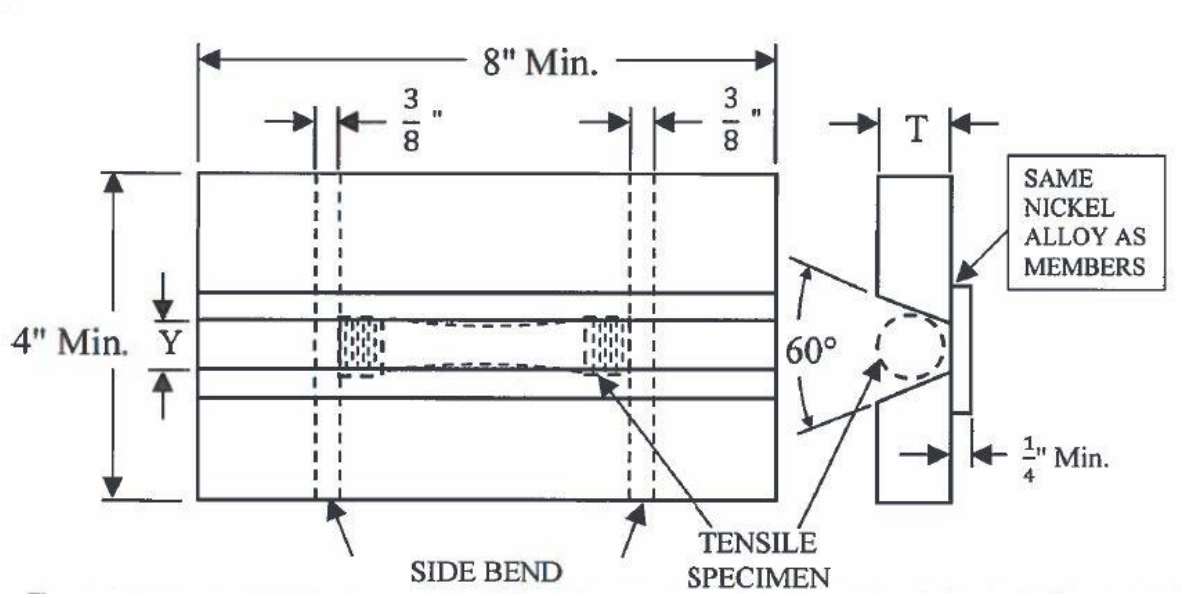
TABLE VII. Inspection levels for determining conformance to cast and helix requirements for rods and electrodes.

Class	Type	Inspection level in accordance with ANSI/ASQ Z1.4 (pound of spools, packages, or drums)
1a	25-pound spools	S-3
1c	100-pound drums	S-3
1d	200-pound drums	S-3
1b	2-pound spools	I
2 (cut rods)	5- or 10-pound packages	S-4
1e to 1i	300- to 700-pound drums	S-2

4.3.1.6 Sampling for chemical analysis. One specimen for chemical analysis shall be selected from each lot (see 4.3.1.1 and 4.5.2.3). If any specimen fails to meet the requirement specified in 3.3.2, this shall be cause for rejection of the lot, subject to the retest requirements of 4.5.3.

4.3.1.7 Sampling for weld tests. Sufficient quantity of class 1 or 2 material, as applicable, shall be selected from each lot (see 4.3.1.1) for preparation of welds and the associated tests performed on test samples removed from the welded test assembly (see figure 1). Failure of any electrode or rod, under test, to meet mechanical property (tensile, side bend), radiographic (RT), or metallurgical (fissures) requirements shall be cause for rejecting the lot, subject to the retest requirements of 4.5.3.

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Class	Diameter Sizes	Y (inch)	T (inch)	Tensile Specimen Diameter (inch)	Welding Process
1a through 1i – wound electrode	All	½ ±1/16	¾ minimum	0.500	Gas metal – arc
2 – cut length rod	All	¼ ±1/16	½ minimum	0.250	Gas tungsten – arc

NOTES:

1. Weld joint members shall be restrained by clamping to a grooved metal back-up plate. Prior to welding, the assembly may be preset so that the welded joint will be within 5 degrees of plane after welding. As an alternate, restraint or a combination of restraint and presetting may be used. A test assembly that is more than 5 degrees out of plane shall be discarded. Straightening of the test assembly is prohibited.
2. See 3.3.4 (Soundness), 4.5.1 (Test weld fabrication), 4.5.2.4.1 (Tensile), 4.5.2.4.2 (Side bends), 4.5.2.5 (RT), and note 3 of table VI.

FIGURE 1. Test joint for mechanical properties.

4.3.1.8 Sampling for containers, container marking, and lot identification examinations. Unless otherwise specified (see 6.2), the examination of each lot for verification of conformance to container, container marking, and lot identification requirements of section 3 shall be in accordance with MIL-W-10430.

4.3.1.9 Sampling for plastic spool burn test. Sampling shall be as specified (see 6.2).

4.3.2 Certification of conformance. Unless otherwise specified (see 6.2), a certificate of conformance (CoC) shall be prepared for each lot of material offered for acceptance in accordance with the lot definition as specified herein. The CoC shall be in accordance with the requirements herein. A CoC or equivalent test certificate (see 4.3.2.9) shall be furnished to the consignee with the shipment and shall meet the following requirements.

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4.3.2.1 Test results. The certificate shall include quantitative results for the specified tests herein. The certification shall also include qualitative results of nondestructive tests and other inspections and tests. The cause for any retest shall be reported for information and the test results that failed to meet those specified requirements shall be reported.

4.3.2.2 Stress relief. A record of any stress relief treatment (if applicable) shall be reported.

4.3.2.3 Age hardening. A record of any age hardening treatment (if applicable) shall be reported.

4.3.2.4 Electrode flux combination. When required to use a specific electrode and flux combination, the certification shall report the brand name of flux used in the conformance testing.

4.3.2.5 Lot information. The heat number, lot number, and material source (melter) shall be reported.

4.3.2.6 Welding information. Welding process and operational parameters for welded test assemblies shall also be reported.

4.3.2.7 Other information. The manufacturer, distributor customer, order number, specification, MIL-type, heat and lot, size, and class shall be identified.

4.3.2.8 Certification statement. The certificate shall state that each lot has been sampled, tested, and inspected in accordance with the specification requirements to produce an electrode or rod, as applicable, that meets the requirements. The certificate shall be signed by a responsible representative of the manufacturer.

4.3.2.9 Test certificates. Where test certificates issued by the manufacturer contain the above information (see 4.3.2 through 4.3.2.8), a separate CoC will not be required.

4.4 Examinations and tests. Samples selected in accordance with 4.2.3 and 4.3.1, as applicable, shall be subject to the examinations and tests for qualification and conformance inspections as designated in tables V and VI. Procedures and methods shall be as specified in 4.5. Retest, when performed, shall be in accordance with 4.5.3.

4.5 Procedures and methods. Procedures and methods shall be in accordance with 4.5.1 through 4.5.3.1.3.

4.5.1 Test weld fabrication. Requirements for test weld fabrication shall be in accordance with 4.5.1.1 through 4.5.1.7.

4.5.1.1 Welding machines. Manual, semi-automatic, or machine equipment shall be used with either the GMAW or GTAW processes for testing electrode or cut-length rod. Equipment shall be supplied with welding current from either a motor generator or rectifier in accordance with NEMA EW 1. For both qualification and conformance testing of electrodes, welding machines used to supply power shall be of the following types: (a) variable voltage direct current (DC) motor-generator type, (b) variable voltage or constant voltage DC rectifier welder, (c) variable voltage or constant voltage DC inverter, (d) alternating current (AC) welding transformer, or (e) an AC inverter, all of which shall be in accordance with NEMA EW 1 or IEC 60974. Multiple process machines are acceptable provided that they are properly set up. The power source shall be appropriate for the process to be tested and shall be of sufficient rating to supply current demanded by the electrode type and size under test. Open circuit voltage shall not exceed 80 volts. For welding using the pulsed gas metal arc process, the model of the power supply shall be recorded. If applicable, the mode or program used for welding shall be recorded.

4.5.1.2 Instrumentation. For GMAW, voltage shall be measured between the wire feeder drive rolls and the work piece. For GTAW, voltage shall be measured between the test plate and a point as near as possible to the torch or torch cable. Welding current and arc voltage measurements shall be made using digital or analog meters. Digital meters may be root mean square (RMS)-type devices provided the digital RMS meters are only used for direct current and non-pulsed arc welding; RMS meters shall not be used where alternating current or pulsed arc welding is performed. Tong meters shall not be used; only calibrated instruments shall be used. Instrument calibration records shall be traceable to calibration performed in accordance with ISO/IEC 17025 or equivalent standard.

4.5.1.3 Shielding gas. The shielding gas shall be argon in accordance with AWS A5.32.

4.5.1.4 Welding parameters. The range of arc current for each type and size of welding wire for each thickness of plate and shielding gas shall be as recommended.

4.5.1.5 Base plate. The base plate shall be of alloy similar in composition to the electrode or rod under test (see ASTM B167 or B168) except MIL-EN6A weld joints shall consist of one member of Ni Cr-Fe alloy, the other member mild steel or corrosion resisting steel of types 300 or 400 series.

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4.5.1.6 Preheat and interpass temperatures. The minimum preheat temperature shall be 60 °F and the maximum interpass temperature shall be 300 °F.

4.5.1.7 Welding position. The welding position shall be flat.

4.5.2 Examination and test details. Procedures and methods for tests and examinations for conformance to the section 3 requirements shall be in accordance with 4.5.2.1 through 4.5.2.8.

4.5.2.1 Pre-production inspection of equipment. Prior to the start of a new production run, where a different electrode type is to be produced, inspection shall be performed to assure that equipment and process lines are purged of material from the previous production run that do not meet the requirements herein or conditions for qualification for the new production run. This inspection shall be performed for all equipment used in manufacturing operations where the material is not segregated and positively identified, for example, cutting to length, extruding, and baking (see 3.2.22).

4.5.2.2 Examinations. Examinations shall be in accordance with 4.5.2.2.1 and 4.5.2.2.2.

4.5.2.2.1 Finish, diameter, length, winding, identification, and containers. The sample lengths and pieces selected in accordance with 4.3.1.3 and 4.3.1.8, as applicable, shall be examined to verify conformance to 3.3.1 (finish), 3.4.3 (winding), 3.4.4 (class 1 identification), 3.5.3 (class 2 identification), and 3.7 (container and container marking) and measured to verify conformance as specified in 3.4.1 (class 1 diameter tolerance), 3.4.2 (weight of wound electrode), 3.5.1 (class 2 diameter tolerance), and 3.5.2 (class 2 length).

4.5.2.2.2 Weight. The weight of unit containers of wound electrode shall be measured to verify compliance as specified in 3.4.2.

4.5.2.3 Chemical analysis. Chemical analysis shall be in accordance with ASTM E354. Other methods are acceptable if the accuracy of the method is demonstrated by comparing the results with a standard sample of similar material traceable to the National Institute of Standards and Technology (NIST) or equivalent. The ladle analysis may be used in lieu of the finished electrode or rod analysis when the following provisions are met (see 3.3.2):

a. The melting of the alloy is a part of the contractor's integrated production systems.

b. The ladle sample is taken from the molten stream while pouring the ladle contents into the ingot mold. The ladle sample is then analyzed to verify conformance to 3.3.2.

c. A statistical study shows that the melting and pouring practice is such that there is no essential difference in the chemical composition of the ladle sample and the finished electrode or rod sample (see 4.3.1.6).

d. The two samples are identified by the same heat number.

e. The ladle sample has the closest spectrographic comparison match with the finished electrode or rod sample of all the heats of the same MIL-type melted in the preceding 12 months.

f. In case of a dispute, the wet chemical analysis of the finished electrode or rod shall be used.

4.5.2.4 Mechanical tests. Mechanical tests shall be in accordance with 4.5.2.4.1 through 4.5.2.4.2.

4.5.2.4.1 Tensile. All weld metal tensile specimens shall be prepared and tested in accordance with AWS B4.0. Tension testing shall be conducted at room temperature (see 3.3.3).

4.5.2.4.1.1 Rounding-off procedures for tensile testing. For the purposes of determining conformance with this specification, an observed or calculated value shall be rounded off to the nearest 1,000 pounds per square inch for tensile and yield strength and to the “nearest unit” in the last right-hand place of figures used in expressing the limiting value for other values in accordance with the rounding-off method specified in ASTM E29.

4.5.2.4.1.2 Check mechanical property tolerances. Unless otherwise specified (see 6.2), retest as part of receipt inspection by a shipyard or other user of a product supplied in conformance with this specification shall use mechanical property tolerances to verify acceptability of the product as follows:

a. Yield strength test results (rounded off in accordance with 4.5.2.4.1.1) may be 3 percent higher than the specified maximum or 3 percent lower than the specified minimum requirements.

b. Ultimate tensile strength test results (when required) may be lower than the specified requirement by 3 percent (rounded off in accordance with 4.5.2.4.1.1).

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c. Elongation test results shall be not lower than 93 percent of the specified minimum requirement.

4.5.2.4.2 Side bends. Side bend specimens shall be prepared and tested in accordance with AWS B4.0 for electrodes and rod types requiring side bend tests. (See table IV herein for electrode and rod types requiring side bend tests.)

4.5.2.5 Radiographic testing (RT)(soundness). Groove welds shall be radiographed in accordance with T9074-AS-GIB-010/271, using sensitivity level 2-2T after removal of backing strip and grinding smoothly the as-welded surface (see 3.3.4).

4.5.2.6 Alloy identity. Alloy identity test methods and procedures shall be in accordance with AWS A5.01M/A5.01 Department of Defense (DoD) supplementary requirements. The test method(s) and procedure(s) shall be submitted for approval to NAVSEA (see 3.3.5).

4.5.2.7 Cast and helix. Verification of cast and helix shall be in accordance with 4.5.2.7.1 through 4.5.2.7.3 (see 3.4.3.1).

4.5.2.7.1 Class 1a. Specimens selected in accordance with 4.3.1.4 shall be tested in accordance with 4.5.2.7.1.1 to determine conformance to 3.4.3.1.1 and 3.4.3.1.2. If any specimen fails either examination, both tests shall be performed on every spool and all failures shall be rejected, subject to the retest requirements of 4.5.3.

4.5.2.7.1.1 Class 1a cast and helix procedure. The cast and helix of spooled and coiled electrodes shall be measured by cutting a length required to impart to the electrode a curvature such that one loop is formed when placed on a flat surface forming an unrestrained circle.

4.5.2.7.2 Class 1b. The verification method for determining compliance with 3.4.3.1.3 shall be in accordance with the manufacturer's written procedures. Measurement of cast and helix is one acceptable verification method when the criteria set by the manufacturer is indicative of acceptable feeding behavior of the electrode in the welding equipment.

4.5.2.7.3 Classes 1c through 1i. Verification method for determining compliance with 3.4.3.1.4 shall be in accordance with the manufacturer's written procedures. Measurement of cast and helix is one acceptable verification method when the criteria set by the manufacturer is indicative of acceptable feeding behavior of the electrode in the welding equipment.

4.5.2.8 Plastic spool burn tests. Plastic spool burn test procedures shall be in accordance with ASTM D635 (see 3.6).

4.5.3 Unsatisfactory test results. Retest shall be in accordance with 4.5.3.1 through 4.5.3.1.3.

4.5.3.1 Retests. When any required original conformance test or examination representing a lot of material fails to meet specification requirements, the lot may be retested at the manufacturer's discretion. Retest is not applicable to pre-production inspection of equipment (see 4.5.2.1) or container, container marking, and lot identification examinations (see 4.5.2.2.1). If the manufacturer chooses to retest the lot, the retests shall be as follows:

a. If the test failed due to one (or more) of the conditions listed in 4.5.3.1.1, then the lot shall be retested according to the requirements in 4.5.3.1.2.

b. If the test failed due to conditions other than those listed in 4.5.3.1.1, then the lot shall be retested according to the requirements in 4.5.3.1.3.

Note that all the specimens in a single test, whether the test involves a single specimen (such as for radiographic examination) or multiple specimens (such as for CVN impact tests), shall be considered a "set".

4.5.3.1.1 Testing or specimen preparation error retests. Retests to correct errors in testing and test specimen preparation are permitted under the following conditions:

a. Improperly fabricated test welds, such as test welds that have not been fabricated in accordance with the applicable specification or the weld procedure for the product under test.

b. Improperly prepared specimens, such as test specimens that are not in compliance with applicable standards.

c. Test equipment malfunction in the process of conducting a test.

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d. Flaws in test specimens that are not indicative of inferior or defective product. Flaws at weld starts and stops at the ends of the test assembly, lack of fusion at the root, and slag due to operator technique are normal weld artifacts and do not count as flaws indicative of inferior or defective products. Cracks, clustered porosity, flakes, and ruptures are flaws that may be indicative of inferior or defective products and shall not qualify a lot for a testing or specimen preparation error retest.

4.5.3.1.2 Testing or specimen preparation error retest requirements. Retests for conditions listed in 4.5.3.1.1 shall be as follows:

a. For each original test that failed due to one (or more) of the conditions in 4.5.3.1.1, only one testing or preparation error retest is permitted.

b. A testing or preparation error retest shall consist of one replacement specimen for each specimen that failed due to the conditions in 4.5.3.1.1. Note for radiographic examination, the specimen is the entire weld. If the testing error is due to a malfunction of magnetic particle testing (MT) or RT equipment or to a flaw not indicative of inferior or defective products (see 4.5.3.1.1d) removed by grinding or other methods within $\frac{1}{16}$ inch of the surface, the original test assembly may be retested.

c. Results of all retests and reasons for all retests shall be reported on the certification of conformance.

d. Retests of the weld metal (e.g., mechanical properties) shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

e. Retests of the product (e.g., tensiles, side bends) shall be from the same sampling as the original test, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

f. If the results of the testing or preparation error retest do not satisfy the applicable requirements, then a conformance retest shall be performed in accordance with 4.5.3.1.3.

4.5.3.1.3 Conformance retest requirements. Retests for failure to meet test requirements shall be as follows:

a. For each original test that failed due to conditions other than those listed in 4.5.3.1.1, only one conformance retest is permitted.

b. A conformance retest shall consist of repeating the test that failed two times (two-for-one). Note that a mechanical property test such as tensile or side bend involves a set of test specimens and, therefore, two full sets of specimens for each test that failed are required under this two-for-one rule. Results of all retests and reasons for all retests shall be reported on the certification of conformance.

c. Retests of weld metal shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

d. Retests of a product shall be from the same sampling as the original test, if feasible. If material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot or production run.

e. Unless the welding procedure was determined to be the root cause of the original test failure, any replacement test weld should employ the same welding procedure as the original test weld. If a new procedure is used to correct the root cause of the original test failure, the retest shall include all tests conducted for the original weld. Two-for-one is only required for the test that failed. All future lots of the material shall use the new procedure. The reason for changing procedure shall be justified to the customer and included on the certification.

f. If the results of any conformance retest do not satisfy the applicable requirements, then the lot shall be rejected.

4.6 Inspection of packaging and marking. Packaging and marking shall be inspected for compliance with the contract (see 5.1, 6.2, and 6.8).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nickel alloy filler materials specified herein mee minimum standards of quality control and chemical composition. The minimum filler material properties obtained when testing to this specification are considered representative of what can be expected when welding, under controlled conditions, base plates of compositions similar to those of the filler metal, where such compositions are available in wrought forms. Since these filler materials are designed for the welding of these materials to themselves, joining clad materials and welding of dissimilar alloys to ordinary steel or corrosion resisting steels of the AISI 300 and 400 series, the mechanical property requirements indicated herein will generally not be suitable for design purposes. In such instances it is recommended that pilot weld tests be made representative of the welding procedures and alloys involved to establish mechanical properties for representative joints contemplated and made under controlled conditions. Current stock of electrodes and rods may be used until depleted; future requisitions for electrodes and rods should refer to the current version of this specification.

6.1.1 Types MIL-EN 52i and MIL-RN52i. Types MIL-EN52i and MIL-RN52i are recommended for welding nickel-chromium-iron alloy 690 (UNS number N06690) to itself, to steels, and to weld overlay steels in nuclear pressure vessel applications. It also shows excellent resistance to primary water stress corrosion cracking in intended service conditions.

6.1.2 Types MIL-EN60 and MIL-RN60. Types MIL-EN60 and MIL-RN60 are recommended for welding nickel-copper alloy to itself and to steel, welding the clad side of nickel-copper alloy clad steel, and for overlaying nickel-copper alloy on steel. For certain overlaying applications, a barrier layer of nickel is required below the nickel-copper alloy overlay, and the manufacturer should be consulted.

6.1.3 Types MIL-EN61 and MIL-RN61. Types MIL-EN61 and MIL-RN61 are recommended for welding of nickel to itself, welding the clad side of nickel clad steel, and overlaying nickel on steel.

6.1.4 Types MIL-EN62 and MIL-RN62. Types MIL-EN62 and MIL-RN62 are recommended for welding nickel-chromium-iron alloy to itself.

6.1.5 Types MIL-EN64 and MIL-RN64. Types MIL-EN64 and MIL-RN64 are recommended for welding nickel-copper-aluminum alloy to itself. The deposited weld metal is age-hardenable, although the increment from age-hardening and mechanical properties to be expected after welding and aging will be lower than values obtainable from wrought material.

6.1.6 Type MIL-RN65. Type MIL-RN65 is recommended for GTAW of nickel-iron-chromium-molybdenum-copper alloy to itself. GMAW of this material is not recommended.

6.1.7 Types MIL-EN67 and MIL-RN67. Types MIL-EN67 and MIL-RN67 are recommended for welding of 70-30, 80-20, and 90-10 copper-nickel alloys to themselves and to steel.

6.1.8 Type MIL-RN69. Type MIL-RN69 is recommended for tungsten-arc welding of nickel-chromium-iron-titanium-aluminum-columbium to itself. The deposited weld metal is age-hardenable, although the increment from age-hardening and mechanical properties to be expected after welding and aging will be lower than values obtainable from wrought material.

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6.1.9 Types MIL-EN6A and MIL-RN6A. Types MIL-EN6A and MIL-RN6A are recommended for welding of nickel-chromium-iron-titanium-manganese alloys to steel, for overlaying nickel-chromium-iron alloy on steel, and for joining various heat and corrosion resistant alloys to each other and to steel and for welding the clad side of nickel-chromium-iron alloy clad steel.

6.1.10 Types MIL-EN6N and MIL-RN6N. Types MIL-EN6N and MIL-RN6N are recommended for joining of 80-20 nickel-chromium alloys to themselves, and for over-laying nickel-chromium alloy on steel.

6.1.11 Types MIL-EN82, MIL-EN82H, MIL-RN82, and MIL-RN82H. Types MIL-EN82, MIL-EN82H, MIL-RN82, and MIL-RN82H are recommended for welding of nickel-chromium-iron alloy to itself and to steel, for overlaying nickel-chromium-iron alloy on steel, and for welding the clad side of Ni-Cr-Fe alloy clad steel.

6.1.12 Types MIL-EN625 and MIL-RN625. Types MIL-EN625 and MIL-RN625 are recommended for welding of nickel-chromium-molybdenum-columbium alloy to itself and to steel and the overlaying of nickel-chromium-molybdenum-columbium alloy on steel.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class, sub-class (spool or drum size), type, and diameter size required (see 1.2).
c. Spooled electrode, drums of electrode, and cut-length rods should be ordered in units as follows:
   (1) Class 1a – 25-pound (wound on spool).
   (2) Class 1b – 40-pound (wound on twenty 2-pound spools).
   (3) Class 1c – 100-pound (furnished in a drum).
   (4) Class 1d – 200-pound (furnished in a drum).
   (5) Class 1e – 300-pound (furnished in a drum).
   (6) Class 1f – 400-pound (furnished in a drum).
   (7) Class 1g – 500-pound (furnished in a drum).
   (8) Class 1h – 600-pound (furnished in a drum).
   (9) Class 1i – 700-pound (furnished in a drum).
   (10) Class 2 (rods) – 5- or 10-pound.
d. Specify the specific issues of the required individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
e. If a minimum carbon content is required (see table III, note 3).
f. If maximum sulfur content is restricted to 0.007 percent (see table III, note 4).
g. Whether alloy identity testing of each electrode or rod after final marking is required (see 3.3.5.2).
h. Tolerances on weight for classes lc through li (see 3.4.2c).
i. Whether 12-, 18-, or 36-inch rods are required (see 3.5.2).
j Whether the plastic spool burn test is required and the sampling required (see 3.6 and 4.3.1.9).
k. Whether containers are required to be in accordance with MIL-W-10430 or AWS A5.02/A5.02M (see 3.7.1).
l. Additional container requirements (see 3.7.1).
m. Additional container marking requirements (see 3.7.3).
n. The percent defective samples (see 4.3.1.5 and 6.7).
o. Whether sampling for containers, container marking, and lot identification examination should be in accordance with requirements other than MIL-W-10430 (see 4.3.1.8).
p. When check mechanical property tolerances are not permitted for retest as part of receipt inspection (4.5.2.4.1.2).
q. Packaging requirements (see 4.6, 5.1, and 6.8).
r. Responsibility for inspection (see 6.4).

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s. Responsibility for compliance (see 6.5).

t. Inspection system requirements (see 6.6).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 21562 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Responsibility for inspection recommendations. Unless otherwise specified (see 6.2) the contractor should be responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Unless otherwise specified (see 6.2), the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

6.5 Responsibility for compliance recommendations. All items should meet all requirements of sections 3 and 4. The inspection set forth in this specification should become a part of the contractor's overall inspection system or quality program (see 6.6). The absence of any inspection requirements in the specification should not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material (see 6.2).

6.6 Inspection system requirement recommendations. The supplier and their contractors should provide and maintain an inspection system acceptable to NAVSEA or its authorized representative for the supplies or services covered by the contract or order. The inspection system should be in accordance with one of the following, unless an equivalent system is approved (see 6.2):

a. ISO 9003 (ANSI/ISO/ASQ 09003) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.

b. ISO 9002 (ANSI/ISO/ASQ 09002) with STR-ISO 9000 Supplement C. Supplement B or A are also acceptable.

c. ISO 9001 (ANSI/ISO/ASQ 09001) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.

d. MIL-I-45208.

6.7 Recommendations for percent defective level for examination of rods and electrodes. It is recommended that an acceptable quality level equal to 2.5 percent defective be utilized for determining the acceptance levels when applying the procedures of ANSI/ASQ Z1.4. This is the same value specified in the previous revisions of this specification (see 4.3.1.5 and 6.2).

6.7.1 Defective material replacement recommendations. When a lot is found to be acceptable when applying any fixed acceptable quality levels, lot tolerance percent defectives, or other type of fixed levels of defects the defective electrodes in the lot found to be defective should be replaced with non-defective electrodes prior to shipment. In no case should known defective material be shipped.

6.8 Packaging recommendations. The following minimum packaging recommendations are provided and are in addition to the container requirements of section 3. Alternate or additional requirements may be necessary as determined by the purchaser.

6.8.1 Protection against damage. Electrodes and rods should be suitably packaged to ensure against damage during shipment and storage under normal conditions.

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6.8.2 Packaging materials. Packaging materials should be in accordance with MIL-W-10430.

6.8.2.1 Loose-fill packaging. The use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation or storage. If cushioning is required, the use of cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material should be specified.

6.8.2.2 Navy fire-retardant materials. Navy fire-retardant requirements for lumber, plywood, and fiberboard used in packaging should be in accordance with MIL-W-10430.

6.9 Recommendations for inspection after delivery. Post-delivery inspection of electrodes, rods, and packaging to determine conformance to this specification and for acceptance thereof, is the responsibility of the purchaser. Post-delivery inspection testing of electrodes intended for critical applications, such as submarine structures, should be in accordance with the applicable fabrication document.

6.10 Mixed-up material recommendations. If, subsequent to shipment, a heat or lot is found to contain material of a type or class different from that specified, the contractor should notify all other customers who procured to the requirements herein and in the applicable specification sheet and received material from that heat or lot of the condition. The contractor should be held responsible for re-inspection to ensure that any unused material from the same heat or lot is correct and should replace all incorrect material at no additional charge.

6.11 Defective material replacement recommendations. When a lot is found to be acceptable when applying any fixed acceptable quality levels, lot tolerance percent defectives, or other types of fixed levels of defects the defective electrodes in the lot found to be defective should be replaced with non-defective electrodes prior to shipment. In no case should known defective material be shipped (see 4.3.1.3.1).

6.12 CoC considerations. The CoC as required in this specification is an essential element for assuring that the material being purchased is in conformance with the specification. The certificate content, as required herein, provides valuable information regarding the welding parameters used for the required tests, expected performance, actual weld metal chemistry, whether the product was subjected to a stress relief heat treatment, information regarding test failures, and other information that links the certificate to the manufacturer, purchaser, specification, lot, etc. Navy technical and safety program documentation typically require certain information included in the prescribed CoC for use as objective quality evidence demonstrating specification conformance and compliance as well as traceability for materials used within prescribed ship systems. Such documentation includes, but is not limited to, SUBSAFE; Level I; Deep Submergence Scope of Certification (DSSOC); and ship hull, machinery, and piping systems fabrication documents. A CoC may also be required by customs. The CoC also provides beneficial information in support of hull structure and component failure analyses; tracking problematic heats or lots of material throughout the supply system or onboard ships; and material, component, and specification improvements. These added benefits are possible because of the unique tracking information provided within the CoC and the unique set of mechanical properties and chemical content provided for each lot of material (see 4.3.2).

6.13 Subject term (key word) listing.

Alloy identity
Gas metal-arc
Gas tungsten-arc
Winding

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Review activity:
DLA - GS

Preparing activity:
Navy – SH
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